Light

GCSE Computer Science — Cyber Security Prevention

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A security system examines network traffic and blocks connections that violate defined rules.
What is this security system called?

- **A.** A compiler
- **B.** A firewall
- **C.** A spreadsheet
- **D.** A web browser

Q2 (1 mark)

Data is transformed into an unreadable format using an algorithm and can only be converted back using a secret key.
What is this process called?

- ○ **A.** Compression
- ○ **B.** Encryption
- ○ **C.** Defragmentation
- ○ **D.** Compilation

Q3 (1 mark)

A user must enter a secret word or phrase known only to them before accessing a computer system.
What is this security measure called?

- ○ **A.** Encryption
- ○ **B.** Defragmentation
- ○ **C.** Compression
- ○ **D.** Password authentication

Q4 (1 mark)

Software scans a computer's files and compares them against a database of known threat patterns.
What is this software doing?

- ○ **A.** Detecting malware using signature-based scanning
- ○ **B.** Compressing files to save storage space
- ○ **C.** Encrypting files for secure transmission
- ○ **D.** Defragmenting the hard drive

Q5 (1 mark)

A company copies all important files to a separate storage location every night.
What is the primary purpose of this practice?

- ○ **A.** To make files load faster
- ○ **B.** To reduce storage costs
- ○ **C.** To enable data recovery if original files are lost, corrupted, or encrypted by ransomware
- ○ **D.** To encrypt the files for security

Q6 (1 mark)

A system administrator configures user accounts so employees can only access files related to their department.
What security concept is being applied?

- ○ **A.** Data compression
- ○ **B.** File defragmentation
- ○ **C.** Network monitoring
- ○ **D.** Access control

Q7 (1 mark)

A data centre uses security guards, CCTV cameras, and electronic door locks.
What type of security measures are these?

- ○ **A.** Software security measures

- ○ **B.** Physical security measures
- ○ **C.** Encryption measures
- ○ **D.** Backup procedures

Q8 (1 mark)

A computer automatically downloads and installs patches from the operating system vendor.
Why is keeping software updated important for security?

- ○ **A.** Updates make the computer screen brighter
- ○ **B.** Updates increase the computer's battery life
- ○ **C.** Updates fix known security vulnerabilities that attackers could exploit
- ○ **D.** Updates delete all user files

Q9 (1 mark)

A security system grants access based on scanning a person's fingerprint or iris.
What type of authentication is this?

- ○ **A.** Password authentication
- ○ **B.** Token authentication
- ○ **C.** Certificate authentication
- ○ **D.** Biometric authentication

Q10 (1 mark)

A document outlines rules for how employees may use company computers and internet access.
What is this document called?

- ○ **A.** An acceptable use policy
- ○ **B.** A software licence
- ○ **C.** A user manual
- ○ **D.** A warranty certificate

Q11 (1 mark)

An employee working from home connects to the company network through an encrypted tunnel over the internet.
What technology creates this secure connection?

- ○ **A.** A search engine
- ○ **B.** A virtual private network (VPN)
- ○ **C.** A spreadsheet application
- ○ **D.** A word processor

Q12 (1 mark)

A company requires all staff to complete annual training on recognising phishing emails and creating strong passwords.
What type of security measure is this?

- ○ **A.** Hardware security

- ○ **B.** Network segmentation
- ○ **C.** Security awareness training
- ○ **D.** Data compression

Q13 (2 marks)

A network administrator configures a firewall that tracks the state of network connections and only allows incoming packets that are part of an established connection.
What type of firewall is this?

- ○ **A.** A stateful firewall that monitors connection states and context, not just individual packets
- ○ **B.** A packet filter that only examines source and destination addresses
- ○ **C.** An antivirus program that scans file contents
- ○ **D.** A backup system that copies data

Q14 (2 marks)

A student learns that one encryption method uses the same key for encryption and decryption, while another uses a pair of different keys. Which statement correctly identifies these methods?

- ○ **A.** Symmetric encryption uses one shared key; asymmetric encryption uses a public key for encryption and a private key for decryption
- ○ **B.** Symmetric encryption requires no keys; asymmetric encryption uses one key
- ○ **C.** Both methods use identical processes with no differences

○ **D.** Symmetric encryption is only for images; asymmetric encryption is only for text

Q15 (2 marks)

A banking app requires customers to enter their password and then confirm a notification on their registered mobile device.
Why does this two-factor authentication provide stronger security than passwords alone?

○ **A.** It makes the password longer and more complex

○ **B.** It encrypts all data stored on the phone

○ **C.** An attacker must compromise both the password (something known) and the phone (something possessed) to gain access

○ **D.** It automatically changes the password every day

Q16 (2 marks)

A company implements the following backup schedule: full backup every Sunday, incremental backups Monday through Saturday.
What is the advantage of this approach compared to daily full backups?

○ **A.** It provides less data protection than full backups alone

○ **B.** It eliminates the need for any backup storage

○ **C.** It automatically encrypts all backed-up data

○ **D.** It uses less storage space and backup time on weekdays because incremental backups only capture changes since the last backup

Q17 (2 marks)

A company places its public web server in a separate network zone between the internet and internal network.
What is this network zone called, and why is it used?

- ○ **A.** A DMZ (demilitarised zone); it isolates public-facing services so that if compromised, attackers cannot directly access the internal network
- ○ **B.** A backup zone; it stores copies of all company data
- ○ **C.** A compression zone; it reduces the size of network traffic
- ○ **D.** A defragmentation zone; it reorganises network packets

Q18 (2 marks)

Anti-malware software detects a suspicious program by analysing its behaviour rather than matching it against known signatures.
What detection method is this?

- ○ **A.** Signature-based detection
- ○ **B.** File compression
- ○ **C.** Heuristic or behavioural analysis that identifies threats by suspicious actions rather than known patterns
- ○ **D.** Data defragmentation

Q19 (2 marks)

An IT department ensures that users can only access the minimum resources necessary for their specific job functions.
What security principle does this implement, and why is it important?

- ○ **A.** Principle of least privilege; it limits damage from compromised accounts and reduces insider threat risk
- ○ **B.** Maximum access principle; users need access to everything for convenience
- ○ **C.** Open access principle; it improves collaboration
- ○ **D.** Universal access principle; it simplifies administration

Q20 (2 marks)

Before disposing of old computers, a company uses software to overwrite all hard drive data multiple times with random patterns. Why is this practice necessary?

- ○ **A.** It makes the computers run faster before disposal
- ○ **B.** Deleted files can be recovered with specialist tools; secure wiping prevents data recovery from disposed equipment
- ○ **C.** It reduces the weight of the computers
- ○ **D.** It improves the resale value of the computers

Q21 (2 marks)

A company hires security professionals to attempt to break into their systems using the same techniques real attackers would use. What is this practice called, and what is its purpose?

- ○ **A.** Data compression; it reduces file sizes
- ○ **B.** System defragmentation; it improves disk performance
- ○ **C.** Load balancing; it distributes network traffic

○ **D.** Penetration testing; it identifies vulnerabilities before malicious attackers can exploit them

Q22 (2 marks)

A company encrypts files stored on their servers and also encrypts data when transmitting it over the network.
What are these two types of encryption called?

○ **A.** Fast encryption and slow encryption

○ **B.** Visible encryption and invisible encryption

○ **C.** Temporary encryption and permanent encryption

○ **D.** Encryption at rest (stored data) and encryption in transit (transmitted data)

Q23 (2 marks)

A company's password policy requires: minimum 12 characters, uppercase and lowercase letters, numbers, symbols, and passwords cannot be reused for 12 months.
How do these requirements improve security?

○ **A.** They make passwords easier to remember

○ **B.** Complexity makes passwords harder to guess or crack; reuse prevention stops compromised passwords from being reused

○ **C.** They reduce the storage space needed for passwords

○ **D.** They eliminate the need for other security measures

Q24 (3 marks)

A security consultant reviews an organisation's defences:
• Firewall blocking unnecessary ports

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• Anti-malware on all endpoints
• Email filtering for phishing
• Regular security awareness training
• Data encryption at rest and in transit
• Daily backups stored off-site
```

What security strategy do these measures represent, and why is this approach effective?

- ○ **A.** Defence in depth; multiple layers ensure that if one control fails or is bypassed, others remain to protect assets — no single point of failure can compromise all security
- ○ **B.** Single point of failure security; if one measure fails, all security is lost
- ○ **C.** Security through obscurity; hiding systems provides all necessary protection
- ○ **D.** Minimal security; fewer controls reduce complexity

Q25 (3 marks)

An organisation evaluates authentication methods for accessing sensitive financial systems:
• Option A: Complex passwords only
• Option B: Passwords plus SMS codes
• Option C: Passwords plus hardware security keys
Which evaluation correctly assesses these options?

- ○ **A.** All three options provide identical security levels
- ○ **B.** Option A is strongest because passwords cannot be compromised

○ **C.** Option C provides the strongest security: hardware keys are resistant to phishing and cannot be intercepted like SMS; Option B improves on passwords but SMS can be compromised through SIM-swapping; Option A is vulnerable to phishing and credential theft

○ **D.** Option B is strongest because everyone has mobile phones

Q26 (3 marks)

A company's backup strategy includes:
• Automated daily backups to a network-attached storage device
• The backup drive is permanently connected to the network
• No off-site or cloud copies exist
• Backups are not encrypted
Ransomware infects the network and encrypts all accessible files including the backup drive.
Which analysis correctly identifies the backup strategy's weaknesses?

○ **A.** Multiple critical weaknesses: permanently connected backup was accessible to ransomware; no off-site copy meant no recovery option remained; lack of encryption means stolen backups could be read; air-gapped or offline backups would have survived

○ **B.** The strategy was perfect; ransomware would have encrypted any backup regardless of configuration

○ **C.** The backup frequency was the only problem; daily backups should be hourly

○ **D.** Backup encryption would have prevented the ransomware infection entirely

Q27 (3 marks)

A network architect designs a school network with:
• Separate network segments for student devices, staff computers, and servers

```
• Firewalls controlling traffic between segments
• Student network cannot directly access the server segment
• Guest WiFi isolated from all internal networks
```

How does this design improve security compared to a flat network where all devices are on the same segment?

- ○ **A.** Segmentation provides no security benefits and only increases costs
- ○ **B.** Segmentation contains breaches: if a student device is compromised, attackers cannot directly reach servers; malware cannot spread freely across the entire network; each segment can have appropriate security policies for its users and purpose
- ○ **C.** Flat networks are more secure because traffic flows more easily
- ○ **D.** Segmentation only affects network speed, not security

Q28 (3 marks)

A company uses strong encryption to protect sensitive data, but:
• Encryption keys are stored in the same database as the encrypted data
• All employees have access to the key storage location
• Keys are never rotated or changed
• No key backup exists
Which analysis correctly identifies the weaknesses in this implementation?

- ○ **A.** This is a perfect encryption implementation with no weaknesses
- ○ **B.** Encryption strength is the only factor that matters; key management is irrelevant
- ○ **C.** Keys should be shared with as many people as possible for convenience
- ○ **D.** Storing keys with data defeats encryption's purpose (attackers who access the database get both); universal key access violates least

privilege; never rotating keys means any compromise is permanent; no key backup risks permanent data loss if keys are corrupted

Q29 (3 marks)

A security manager proposes a comprehensive security program including:
• Technical controls (firewalls, encryption, anti-malware)
• Administrative controls (policies, procedures, training)
• Physical controls (locks, cameras, access cards)
• Regular audits and penetration testing
• Incident response planning
Why does effective security require all these components rather than focusing on just technical controls?

○ **A.** Technical controls alone provide complete protection; other components are unnecessary

○ **B.** Administrative controls alone are sufficient; technology is unnecessary

○ **C.** Security threats target technical, human, and physical vulnerabilities; comprehensive defence requires controls addressing each area; policies guide behaviour, training reduces human error, physical security prevents direct access, and testing validates effectiveness

○ **D.** Physical controls alone are sufficient; cyber threats can be ignored

Q30 (3 marks)

After a security breach, a company's post-incident review reveals:
• The attack exploited a vulnerability patched 6 months earlier
• Security alerts were generated but ignored
• No incident response plan existed
• Backups had not been tested and failed during recovery
Which preventive measures would have addressed these failures?

- ○ **A.** Faster internet connections would have prevented the breach
- ○ **B.** More encryption would have solved all these problems
- ○ **C.** These failures cannot be prevented through any measures
- ○ **D.** Timely patch management would have closed the vulnerability; security monitoring with alert response procedures would have detected the attack; a tested incident response plan would have guided recovery; regular backup testing would have ensured recovery capability

Total: **55** marks • Time: **30 min**

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